Trading Competition Report

BUS 350 – Investment Analysis
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1. Introduction

When the trading competition began, my objective was to build a portfolio that I would realistically feel comfortable holding in real life. Although the simulation provided one million dollars, I approached the task as if it represented my own long-term savings. From the start, my focus was on developing a passive, long-term portfolio oriented toward low to moderate risk and steady, sustainable returns rather than short-term speculation.

My intention was to create a diversified structure that could withstand volatility and gradually compound over time. Initially, I considered expanding across multiple industries—technology, retail, healthcare, consumer staples, and potentially energy. However, diversification only adds value when the selected securities provide attractive risk-return characteristics. After reviewing several potential candidates, I chose four assets that demonstrated strong long-term qualities: Apple (AAPL), Costco (COST), Nvidia (NVDA), and IVV (the iShares Core S&P 500 ETF). The combination gave me balanced exposure to growth, defensiveness, and market-wide performance.

Throughout the analysis, I used a five-year data window from October 1, 2020, to October 1, 2025, to evaluate risk and return profiles.

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2. Research Process and Analytical Framework

To approach this assignment as an analyst, I based decisions on quantitative evidence supported by qualitative reasoning. For each security, I collected daily adjusted closing prices over the five-year period and calculated several key performance metrics:

- Daily percentage returns
- Log returns
- Annualized return
- Annualized volatility
- Sharpe and Sortino ratios
- Beta relative to the market
- CAPM expected return
- Alpha (excess return beyond CAPM)
- Risk-to-return efficiency

These metrics helped form a clear picture of each stock's performance, risk behavior, and suitability for a passive long-term portfolio.

One of the most informative tools in the analysis was the Risk-to-Return Scatterplot, which mapped annualized return against annualized volatility. A trendline was added to represent the average risk-return relationship among all evaluated stocks. Securities positioned above this line offered strong returns relative to their volatility. Those below the line underperformed on a risk-adjusted basis.

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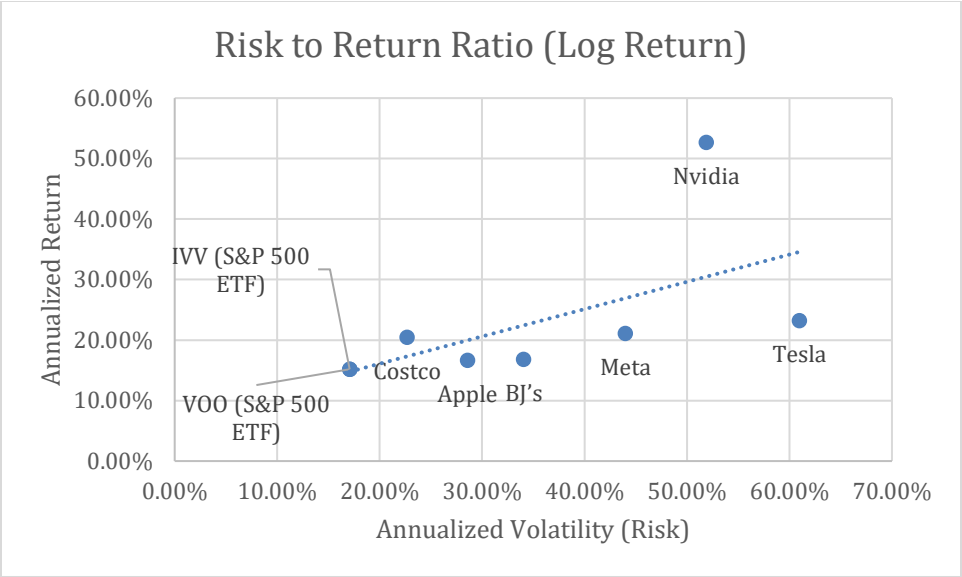
3. Selection of the Four Securities & Expanded Risk-to-Return Chart Interpretation

Although the assignment required selecting at least four securities, I initially intended to diversify more broadly. However, after evaluating several healthcare and retail alternatives, many did not meet my long-term risk-return standards. Rather than include sectors for the sake of diversification, I chose the four assets that best aligned with the portfolio's goals of stability, efficiency, and moderate growth.

Apple, Costco, Nvidia, and IVV represented a high-quality blend of consistent growth, defensiveness, and market-wide exposure.



The Risk-to-Return Scatterplot served as the visual backbone of my entire selection process. Plotting annualized return on the vertical axis and annualized volatility on the horizontal axis allowed me to see immediately which stocks were turning risk into reward efficiently. The trendline on the chart acted as the "expected line," showing what an average stock's return should look like given a certain level of volatility.

Nvidia sat far above and to the right of the trendline. Its volatility is undoubtedly high, but its return was so strong that it more than justified the additional risk. This placed Nvidia firmly in the high-risk, high-reward quadrant, signaling that although it could contribute volatility, it was contributing even more performance.

Costco, on the other hand, sat clearly above the trendline but in a lower-risk area of the chart. This indicated exceptional efficiency. For each unit of volatility Costco produced, it delivered more than one full unit of return. This made it one of the best risk-adjusted performers in the entire dataset.

Apple sat almost exactly on the trendline. That communicated a balanced, predictable profile—neither overperforming nor underperforming relative to its level of volatility. Its position suggested that Apple was a steady, reliable contributor with a risk-return profile that aligned with broad expectations.

IVV appeared in the lower-left quadrant, representing low risk and stable moderate returns. Its position reinforced its role as a stabilizer. While it did not outperform the market dramatically, it provided consistency, anchoring the portfolio and reducing overall volatility.

Seeing these four positions together on the scatterplot helped confirm that the chosen portfolio was well-structured. Costco and Nvidia offered strong risk-adjusted performance, Apple added steady growth, and IVV grounded the portfolio with market-wide exposure.

Stock	Return	Risk	Sharpe	Sortino	Beta	Alpha	Return/Risk
Nvidia	66%	52%	1.18	1.89	2.12	36%	1.26
Costco	23%	22.60%	0.82	1.07	0.78	9%	1.02
VOO / IVV	16.60%	~17%	0.71	0.98	1	0%	0.98
Meta	30.70%	43.60%	0.6	0.8	0.02	25%	0.7
Apple	20.70%	28.70%	0.57	0.84	-0.04	17%	0.72
Tesla	41.80%	61.25%	0.61	0.96	-0.02	37%	0.68
BJ's	22.60%	34%	0.53	0.72	-0.09	19%	0.66

4. Security Analysis

Stock	Annualized Return	Annualized Risk	Sharpe	Sortino	Beta	Alpha
Apple	20.70%	28.70%	0.57	0.84	-0.04	17%
Costco	23%	22.60%	0.82	1.07	0.78	9%
Nvidia	66%	52%	1.18	1.89	2.12	36%
IVV	16.60%	17%	0.71	0.98	1	0%

4.1 Apple (AAPL)

Apple offered a balanced risk-return profile positioned along the trendline in the scatterplot. With an annualized return of approximately 20.75% and annualized volatility of about 28.67%, Apple demonstrated strong long-term performance without excessive risk. Qualitatively, Apple remains one of the most reliable companies in the world, with a durable ecosystem and predictable demand. The stock's stability and consistent cash flow made it an ideal choice for a long-term passive portfolio.

4.2 Costco (COST)

Costco was one of the strongest performers in the analysis. With an annualized return of roughly 23.03% and annualized volatility around 22.58%, Costco produced a risk-to-return ratio above 1.0, indicating excellent efficiency. Beyond the numbers, Costco's business model—driven by memberships, low turnover, and customer loyalty—provides steady long-term value. Its ethical reputation, worker treatment, and customer-centric approach further reinforce confidence in its defensiveness and consistency.

4.3 Nvidia (NVDA)

Nvidia stood out as the highest-returning asset in the dataset, with annualized returns of approximately 66.25% and volatility near 52.27%. Although highly volatile, Nvidia generated the strongest Sharpe and Sortino ratios among all evaluated securities. Its leadership in artificial intelligence, data centers, and GPU technology justified its inclusion despite its elevated risk. Nvidia served as the portfolio's primary growth engine.

4.4 IVV (iShares Core S&P 500 ETF)

IVV provided broad market exposure with relatively low volatility. Its annualized return of approximately 16.63% combined with volatility near 17% made it an excellent stabilizing component within the portfolio. IVV served as an anchor, reducing concentration risk and ensuring exposure to the overall U.S. economy. The data for IVV was also clean and consistent, making it a reliable benchmark ETF for analysis.

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5. Portfolio Construction and Execution

All trades were executed on October 1, 2025. The intention was to allocate approximately 25% to each of the four securities; due to price differences, the final allocation was slightly varied—28%, 25%, 24%, and 23%—but effectively reflected equal-weighting.

The resulting structure offered:

• Growth potential from Nvidia

• Defensive stability from Costco

• Consistent technological exposure from Apple

• Broad diversification and risk reduction from IVV

This allocation aligned fully with the portfolio's long-term, passive objectives.

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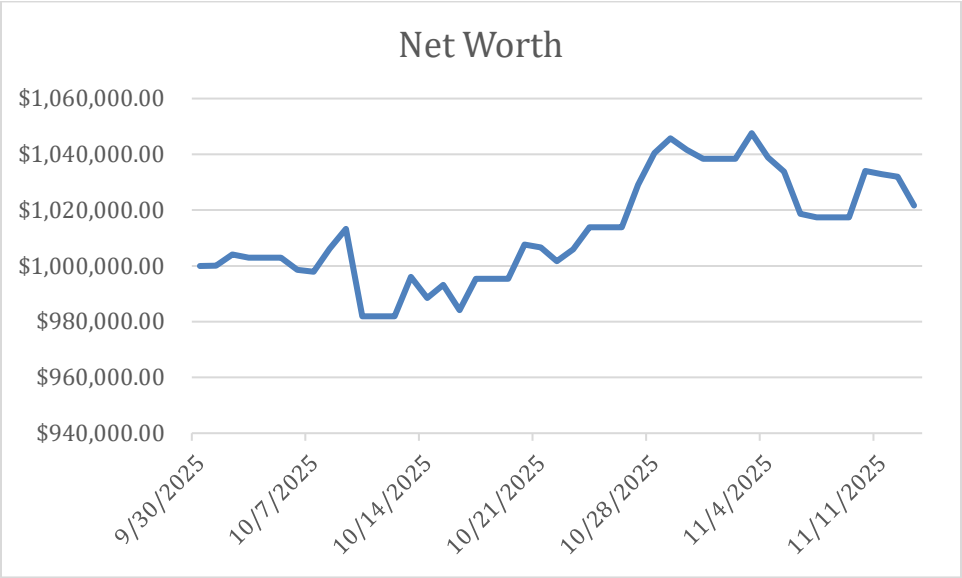
6. Trading Competition Results



I monitored the portfolio approximately every two to three days, consistent with a passive long-term strategy. The final results were:

• Final Net Worth: $1,021,597.10

• Total Return: +2.16%

• Total Gains: $21,597.10

• Final Rank: 18th place

• Highest Rank Achieved: 13th

The portfolio's performance curve reflected a realistic long-term pattern: initial volatility, gradual recovery, steady gains, and temporary pullbacks. The disciplined decision not to sell during volatility proved beneficial, as the portfolio recovered and strengthened over time.

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7. Strategy Evaluation

Overall, the strategy was moderately successful and aligned with the intended long-term approach. The portfolio demonstrated resilience, steady growth, and strong performance relative to its risk profile. Key strengths included:

• High-quality stock selection

• Balanced risk exposure

• Effective use of defensive and growth assets

• Emotional discipline and patience

Potential improvements for future portfolios include expanding sector diversification—particularly into healthcare or industrials—and periodically rebalancing if certain positions grow disproportionately large.

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8. Conclusion

This project provided valuable experience in constructing a disciplined, long-term investment portfolio based on quantitative analysis and real-world reasoning. Through evaluating risk, return, and efficiency, I developed a portfolio that reflects both analytical discipline and practical investment judgment. The experience reinforced the importance of patience, data-driven decision making, and maintaining a long-term perspective in portfolio management.